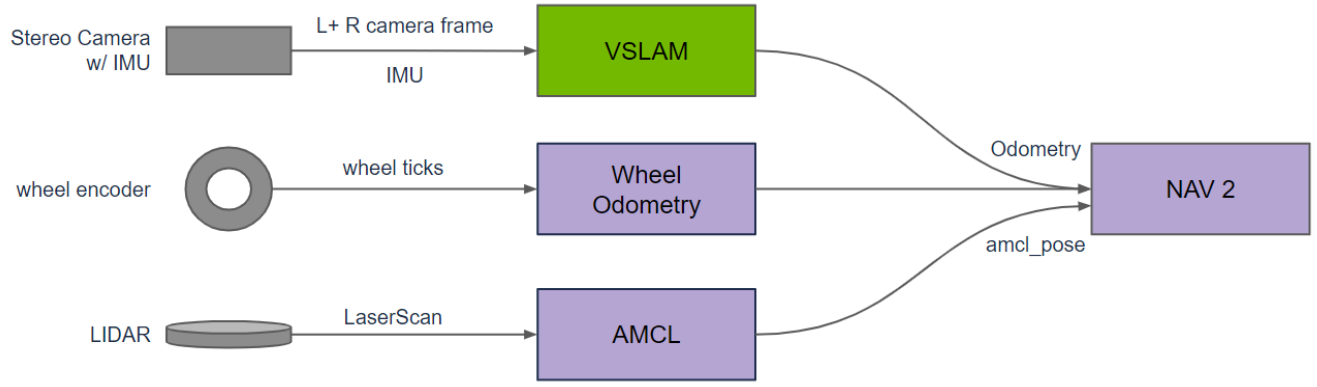
Isaac ROS Visual SLAM

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- Overview
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Isaac ROS Visual SLAM official documentation: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_visual_slam/index.html

Overview

Isaac ROS Visual SLAM uses one or more stereo cameras and an optional IMU to estimate the robot odometry, and can serve as the input for mobile robot navigation or drone localization.



Quick Experience

First start the container with the following command:

```
./run_isaac.sh
```

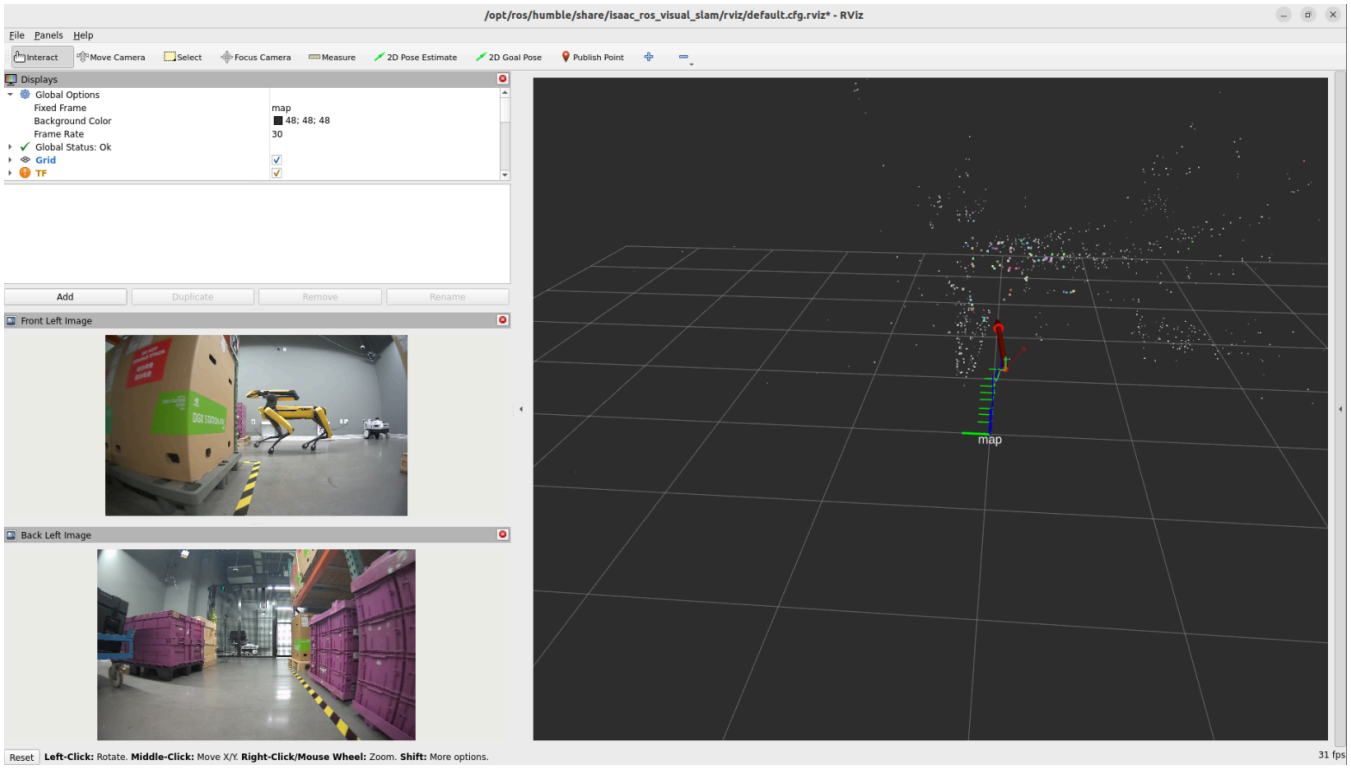
Then confirm the container on the host with `docker ps`, and open container terminals with `docker exec -it <container name> bash`. Every terminal should use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
     STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: All the terminals below run inside the container.

Terminal A: Loop the bag and remap the front stereo camera to the VSLAM inputs:

```
ros2 bag play /workspaces/isaac_ros-
dev/isaac_ros_assets/isaac_ros_visual_slam/quickstart_bag \
  --loop --rate 0.5 --remap \
  /front_stereo_camera/left/image_raw:=/visual_slam/image_0 \
  /front_stereo_camera/left/camera_info:=/visual_slam/camera_info_0 \
  /front_stereo_camera/right/image_raw:=/visual_slam/image_1 \
  /front_stereo_camera/right/camera_info:=/visual_slam/camera_info_1
```



Terminal B: Monitor the odometry:

```
ros2 topic hz /visual_slam/tracking/odometry
```

A single odometry message and the map output topics can also be checked:

```
ros2 topic echo /visual_slam/tracking/odometry --once
ros2 topic info /visual_slam/vis/landmarks_cloud -v
ros2 topic info /visual_slam/tracking/slam_path -v
```

Terminal C: Start the customized cuVSLAM and RViz launch:

```
ros2 launch /workspaces/isaac_ros-dev/launch/vslam.launch.py
```

RViz uses `/workspaces/isaac_ros-dev/rviz/vslam_nano.rviz` with `Fixed Frame` set to `map`. It displays the front-left image `/visual_slam/image_0`, the rear-left image `/back_stereo_camera/left/image_raw`, the SLAM landmarks/path, and single-frame odometry. Do not switch to `vslam_keepall.cfg.rviz`, which overlays a large number of historical odometry graphics.